

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Introduction:**

The purpose of this project is to design and implement a data integration and aggregation process for customer and order data stored in CSV files. The process involves loading the data into Delta tables, performing data cleansing and transformation, and creating aggregated summaries of customer spending.

- Functional Requirements:**

1. **Data Ingestion:**

- Read source CSV data from a specified volume location (`/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``) into Delta tables for customer and order data.
- The schema for the customer table is: ``CustId`, `Name`, `EmailId`, `Region``.
- The schema for the order table is: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``.

2. **Data Cleansing:**

- Remove "Null"/null and duplicate records from both the customer and order tables.

3. **Data Transformation:**

- Create an "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" with the following schema: ``CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date``.
- Join customer and order data using the "CustId" field and load the data into an SCD type 2 table ("ordersummary").

4. **SCD Type 2 Implementation:**

- Implement SCD type 2 logic to update the "ordersummary" table whenever there is a change in the customer table.
- Old records should be made inactive, and new records should be made active.
- Update ``StartDate`` and ``EndDate`` accordingly.

5. **Aggregated Reporting:**

- Create a new table "customeraggregatespend" with columns "Name", "TotalAmount", and "Date" in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist.
- Aggregate the "TotalAmount" column from the "ordersummary" table and group by "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.

- Non-Functional Requirements:**

1. **Data Quality:** Ensure data accuracy, completeness, and consistency throughout the data integration and aggregation process.
2. **Performance:** Design the process to handle large volumes of data efficiently.
3. **Scalability:** Ensure the solution can scale to accommodate growing data volumes and changing business needs.

- Assumptions and Dependencies:**

1. The source CSV files are in the specified location and format.
2. The necessary infrastructure and tools (e.g., Databricks, Delta tables) are available and configured correctly.

- Acceptance Criteria:**

1. The data is successfully ingested into Delta tables.
2. The data is cleansed and transformed as required.
3. The SCD type 2 logic is implemented correctly.
4. The aggregated data is accurately calculated and loaded into the "customeraggregatespend" table.

- Success Metrics:**

1. Data quality metrics (e.g., accuracy, completeness).
2. Process performance metrics (e.g., execution time, resource utilization).

By following this BRD, the project team should be able to design and implement a robust data integration and aggregation process that meets the business requirements.